

HelixTrack E2E Bidirectional Sync — Evidence

Test run: 2026-06-21T05:59:09Z **API:** http://localhost:8080/do **Local DB:** docs/workable_items.db **Status:** PASS (11/11 checks passed)

Summary

Check	Result	Detail
Dependency check	PASS	sqlite3, curl, jq all available
API reachability	PASS	HelixTrack API v1.0.0 at http://localhost:8080/do
JWT acquisition	PASS	Token obtained (349 chars)
DB pre-count	PASS	16 items in local DB before sync
Push sync	PASS	16 items pushed (0 failures)
Ticket list	PASS	96 total tickets in HelixTrack API
Pull sync	PASS	0 added, 96 updated
Count match	PASS	Pre-sync: 16, Post-sync: 16 — identical
ID set match	PASS	OTA-ID set identical before and after sync
Cross-surface OTA coverage	PASS	All 16 local OTA IDs present in HelixTrack (16 unique OTA IDs across 96 total tickets)
Cross-surface ID set equality	PASS	HT unique OTA IDs (16) = Local DB items (16)

DB Items (Post-Sync)

OTA-003|Task|Operator-blocked|Emulator Tier-2 – real Android A/B update_engine/AVB/dm-verity auto-rollback is host-gated
OTA-004|Task|Operator-blocked|Emulator Tier-3 – real RK3588 / Orange Pi 5 Max vendor HAL, U-Boot slot-switch, dm-verity on real

partitions
 OTA-005|Feature|Fixed (→ Fixed.md)|Per-device telemetry filters (?
 event/?since/?until)
 OTA-006|Feature|Fixed (→ Fixed.md)|Group + group-members cursor
 pagination
 OTA-007|Task|Fixed (→ Fixed.md)|Dashboard API client synced to new
 pagination/filter params
 OTA-008|Feature|Fixed (→ Fixed.md)|Tier-1 Go OTA device-emulator
 OTA-009|Task|Fixed (→ Fixed.md)|Comprehensive dashboard UI testing
 system
 OTA-010|Task|Fixed (→ Fixed.md)|Autonomous e2e + security +
 HelixQA test suites
 OTA-014|Task|Fixed (→ Fixed.md)|Docs Chain submodule distribution
 (constitution §11.4.106 Phase 6) operator-gated
 OTA-015|Feature|Fixed (→ Fixed.md)|A/B slot switch via U-Boot
 BOOT_ORDER
 OTA-016|Feature|Fixed (→ Fixed.md)|RAUC dd-apply to inactive slot
 with dm-verity
 OTA-017|Feature|Fixed (→ Fixed.md)|U-Boot corrupt-slot auto-
 rollback via bootcount
 OTA-018|Feature|Fixed (→ Fixed.md)|ApplyPort CLI + slot manager +
 Ed25519 verifier
 OTA-019|Task|Fixed (→ Fixed.md)|Build resource stats tracker
 OTA-020|Bug|Fixed (→ Fixed.md)|Database migration test
 OTA-021|Task|In progress|HelixTrack bidirectional sync
 verification

HelixTrack API Tickets (Post-Push)

Ticket count: 96

Unique OTA IDs: OTA-003 OTA-004 OTA-005 OTA-006 OTA-007 OTA-008
 OTA-009 OTA-010 OTA-014 OTA-015 OTA-016 OTA-017 OTA-018 OTA-019
 OTA-020 OTA-021

Pre/Post Counts

Metric	Value
Pre-sync items	16
Post-sync items	16
HelixTrack total tickets	96
HelixTrack unique OTA IDs	OTA-003 OTA-004 OTA-005 OTA-006 OTA-007 OTA-008 OTA-009 OTA-010 OTA-014 OTA-015 OTA-016 OTA-017 OTA-018 OTA-019 OTA-020 OTA-021
Pre-sync OTA IDs	OTA-003 OTA-004 OTA-005 OTA-006 OTA-007 OTA-008 OTA-009 OTA-010 OTA-014

Metric	Value
Post-sync OTA IDs	OTA-015 OTA-016 OTA-017
	OTA-018 OTA-019 OTA-020
	OTA-021
	OTA-003 OTA-004 OTA-005
	OTA-006 OTA-007 OTA-008
	OTA-009 OTA-010 OTA-014
	OTA-015 OTA-016 OTA-017
	OTA-018 OTA-019 OTA-020
	OTA-021

Sync Output

Push

```
[0;36m[helixtrack-push] [0m Pushing workable items to
HelixTrack...
[0;32m[helixtrack-push] [0m Pushed [OTA-020] Database migration
test
[0;32m[helixtrack-push] [0m Pushed [OTA-019] Build resource stats
tracker
[0;32m[helixtrack-push] [0m Pushed [OTA-003] Emulator Tier-2 –
real Android A/B update_engine/AVB/dm-verity auto-rollback is
host-gated
[0;32m[helixtrack-push] [0m Pushed [OTA-004] Emulator Tier-3 –
real RK3588 / Orange Pi 5 Max vendor HAL, U-Boot slot-switch, dm-
verity on real partitions
[0;32m[helixtrack-push] [0m Pushed [OTA-014] Docs Chain submodule
distribution (constitution §11.4.106 Phase 6) operator-gated
[0;32m[helixtrack-push] [0m Pushed [OTA-005] Per-device telemetry
filters (?event/?since/?until)
[0;32m[helixtrack-push] [0m Pushed [OTA-006] Group + group-
members cursor pagination
[0;32m[helixtrack-push] [0m Pushed [OTA-007] Dashboard API client
synced to new pagination/filter params
[0;32m[helixtrack-push] [0m Pushed [OTA-008] Tier-1 Go OTA
device-emulator
[0;32m[helixtrack-push] [0m Pushed [OTA-009] Comprehensive
dashboard UI testing system
[0;32m[helixtrack-push] [0m Pushed [OTA-010] Autonomous e2e +
security + HelixQA test suites
[0;32m[helixtrack-push] [0m Pushed [OTA-015] A/B slot switch via
U-Boot BOOT_ORDER
[0;32m[helixtrack-push] [0m Pushed [OTA-016] RAUC dd-apply to
inactive slot with dm-verity
[0;32m[helixtrack-push] [0m Pushed [OTA-017] U-Boot corrupt-slot
auto-rollback via bootcount
[0;32m[helixtrack-push] [0m Pushed [OTA-018] ApplyPort CLI + slot
manager + Ed25519 verifier
[0;32m[helixtrack-push] [0m Pushed [OTA-021] HelixTrack
bidirectional sync verification
```

```
[0;36m[helixtrack-push] [0m Sync state written to docs/  
helixtrack_sync_state.md  
[0;32m[helixtrack-push] [0m Push complete – 16 items synced
```

Pull

```
[0;36m[helixtrack-pull] [0m Pulling updates from HelixTrack...  
[0;32m[helixtrack-pull] [0m Fetched 96 tickets from HelixTrack  
[0;36m[helixtrack-pull] [0m UPDATE OTA-021 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-020 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-019 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-003 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-004 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-014 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-005 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-006 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-007 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-008 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-009 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-010 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-015 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-016 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-017 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-018 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-020 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-019 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-003 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-004 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-014 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-005 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-006 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-007 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-008 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-009 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-010 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-015 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-016 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-017 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-018 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-021 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-020 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-019 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-003 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-004 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-014 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-005 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-006 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-007 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-008 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-009 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-010 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-015 → Queued  
[0;36m[helixtrack-pull] [0m UPDATE OTA-016 → Queued
```

```

[0;36m[helixtrack-pull] [0m UPDATE OTA-017 → Queued
[0;36m[helixtrack-pull] [0m UPDATE OTA-018 → Queued
[0;36m[helixtrack-pull] [0m UPDATE OTA-021 → Queued
[0;36m[helixtrack-pull] [0m UPDATE OTA-020 → Queued
[0;36m[helixtrack-pull] [0m UPDATE OTA-019 → Queued
[0;36m[helixtrack-pull] [0m UPDATE OTA-003 → Queued
[0;36m[helixtrack-pull] [0m UPDATE OTA-004 → Queued
[0;36m[helixtrack-pull] [0m UPDATE OTA-014 → Queued
[0;36m[helixtrack-pull] [0m UPDATE OTA-005 → Queued
[0;36m[helixtrack-pull] [0m UPDATE OTA-006 → Queued
[0;36m[helixtrack-pull] [0m UPDATE OTA-007 → Queued
[0;36m[helixtrack-pull] [0m UPDATE OTA-008 → Queued
[0;36m[helixtrack-pull] [0m UPDATE OTA-009 → Queued
[0;36m[helixtrack-pull] [0m UPDATE OTA-010 → Queued
[0;36m[helixtrack-pull] [0m UPDATE OTA-015 → Queued
[0;36m[helixtrack-pull] [0m UPDATE OTA-016 → Queued
[0;36m[helixtrack-pull] [0m UPDATE OTA-017 → Queued
[0;36m[helixtrack-pull] [0m UPDATE OTA-018 → Queued
[0;36m[helixtrack-pull] [0m UPDATE OTA-021 → Queued
[0;36m[helixtrack-pull] [0m UPDATE OTA-021 → Queued
[0;36m[helixtrack-pull] [0m UPDATE OTA-020 → Queued
[0;36m[helixtrack-pull] [0m UPDATE OTA-019 → Queued
[0;36m[helixtrack-pull] [0m UPDATE OTA-003 → Queued
[0;36m[helixtrack-pull] [0m UPDATE OTA-004 → Queued
[0;36m[helixtrack-pull] [0m UPDATE OTA-014 → Queued
[0;36m[helixtrack-pull] [0m UPDATE OTA-005 → Queued
[0;36m[helixtrack-pull] [0m UPDATE OTA-006 → Queued
[0;36m[helixtrack-pull] [0m UPDATE OTA-007 → Queued
[0;36m[helixtrack-pull] [0m UPDATE OTA-008 → Queued
[0;36m[helixtrack-pull] [0m UPDATE OTA-009 → Queued
[0;36m[helixtrack-pull] [0m UPDATE OTA-010 → Queued
[0;36m[helixtrack-pull] [0m UPDATE OTA-015 → Queued
[0;36m[helixtrack-pull] [0m UPDATE OTA-016 → Queued
[0;36m[helixtrack-pull] [0m UPDATE OTA-017 → Queued
[0;36m[helixtrack-pull] [0m UPDATE OTA-018 → Queued
[0;36m[helixtrack-pull] [0m UPDATE OTA-015 → Fixed (→ Fixed.md)
[0;36m[helixtrack-pull] [0m UPDATE OTA-016 → Fixed (→ Fixed.md)
[0;36m[helixtrack-pull] [0m UPDATE OTA-017 → Fixed (→ Fixed.md)
[0;36m[helixtrack-pull] [0m UPDATE OTA-018 → Fixed (→ Fixed.md)
[0;36m[helixtrack-pull] [0m UPDATE OTA-021 → In progress
[0;36m[helixtrack-pull] [0m UPDATE OTA-020 → Fixed (→ Fixed.md)
[0;36m[helixtrack-pull] [0m UPDATE OTA-019 → Fixed (→ Fixed.md)
[0;36m[helixtrack-pull] [0m UPDATE OTA-003 → Operator-blocked
[0;36m[helixtrack-pull] [0m UPDATE OTA-004 → Operator-blocked
[0;36m[helixtrack-pull] [0m UPDATE OTA-014 → Fixed (→ Fixed.md)
[0;36m[helixtrack-pull] [0m UPDATE OTA-005 → Fixed (→ Fixed.md)
[0;36m[helixtrack-pull] [0m UPDATE OTA-006 → Fixed (→ Fixed.md)
[0;36m[helixtrack-pull] [0m UPDATE OTA-007 → Fixed (→ Fixed.md)
[0;36m[helixtrack-pull] [0m UPDATE OTA-008 → Fixed (→ Fixed.md)
[0;36m[helixtrack-pull] [0m UPDATE OTA-009 → Fixed (→ Fixed.md)
[0;36m[helixtrack-pull] [0m UPDATE OTA-010 → Fixed (→ Fixed.md)
[0;32m[helixtrack-pull] [0m Pull sync complete: +0 added, 96
updated, 0 skipped

```

[0;32m[helixtrack-pull] [0m Sync state written to docs/
helixtrack_sync_state.md

Generated by tests/helixqa/helix_e2e_bidir_sync_test.sh at 2026-06-21T05:59:09Z